ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 1ST YEAR

ENGINEERING DRAWING AND CAD

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BEng Mechatronics and Robotics Engineering, 1st year, semester 1 final examination: drawing standards and true lengths, orthographic projection, sectional views and dimensioning, limits and fits, basic manufacturing processes, computer-aided design.

Written by Academic Central for practice, following first-year Engineering Drawing and CAD (graphics) syllabuses at South African universities, following SANS 10111 engineering drawing practice. It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Answer QUESTION 2 on an A3 drawing sheet (or on the grid paper provided), in pencil, using drawing instruments. Answer the other questions in the answer book.
3. Use first-angle orthographic projection unless the question says otherwise.
4. All dimensions are in millimetres.
5. Number your answers exactly as the questions are numbered.

QUESTION 1: DRAWING STANDARDS

1.1 Name FOUR types of line used on engineering drawings and give the use of each. (4)

1.2 A straight edge AB is 40 mm long in the top view, and end B is 30 mm higher than end A. Calculate the true length of AB, and explain how the true length of a line is found by drawing. (3)

1.3 Answer the following:

1.3.1 A part 240 mm long is drawn to a scale of 1:5. How long is it on the drawing? (1)

1.3.2 A part 15 mm long is drawn to a scale of 2:1. How long is it on the drawing? (1)

1.4 Name FOUR items of information that must appear in the title block of a drawing. (4)

1.5 Explain how first-angle and third-angle projection differ in where the top view and the side views are placed. (2)

[15]

QUESTION 2: ORTHOGRAPHIC PROJECTION

FIGURE 1 shows an isometric view of a stepped support block. The base is 80 long, 40 deep and 20 high. The upright at the left end is 30 long, 40 deep and rises to a total height of 50. A Ø16 hole goes vertically through the base; its centre is 55 from the left end and 20 from the front.

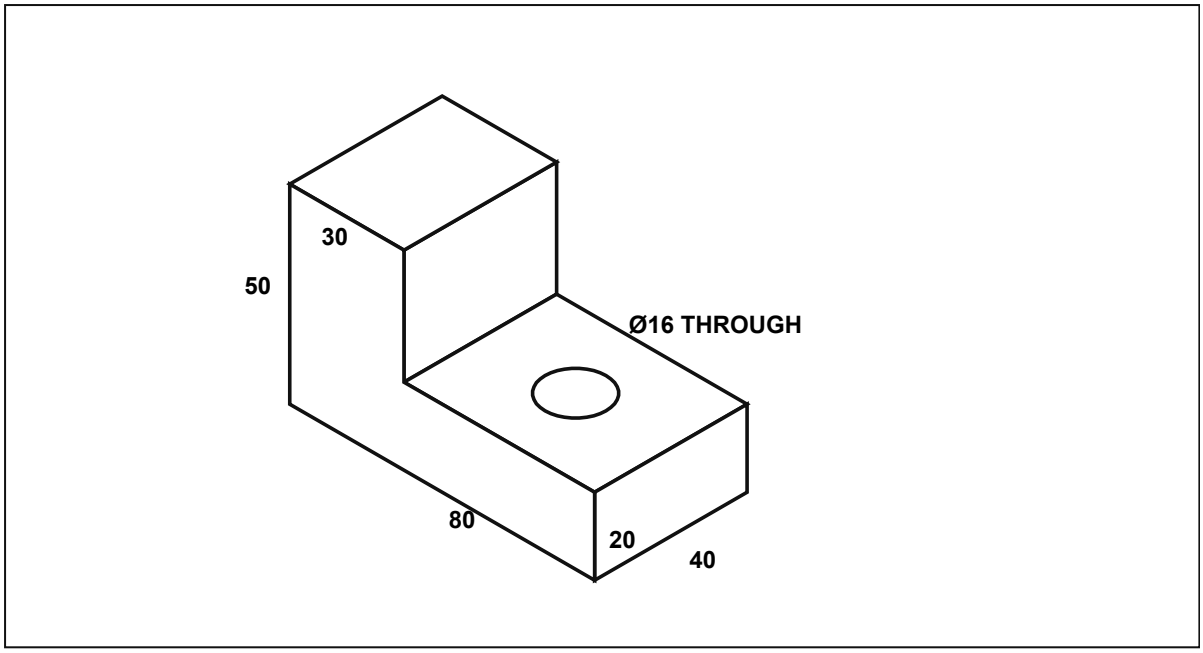


FIGURE 1: Stepped support block (isometric view, not to scale)

2.1 Draw, to a scale of 1:1 and in first-angle projection, the following views of the block:

2.1.1 the front view (looking at the face with the L shape) (8)

2.1.2 the top view, placed correctly in relation to the front view (8)

2.1.3 the left view, placed correctly in relation to the front view (6)

2.2 Show all hidden detail and centre lines. (4)

2.3 Add the first-angle projection symbol and FOUR main dimensions. (4)

[30]

QUESTION 3: SECTIONAL VIEWS AND DIMENSIONING

3.1 Why are sectional views used? Give TWO rules for hatching a sectioned part. (4)

3.2 Name THREE parts that are normally NOT hatched when the cutting plane passes along their length. (3)

3.3 Give FIVE rules for dimensioning a drawing correctly. (5)

3.4 Explain the thread note M12 × 1.75. (3)

3.5 Explain the difference between chain dimensioning and parallel (datum) dimensioning, and say which one gives better accuracy and why. (3)

3.6 What is the difference between aligned and unidirectional dimensions? (2)

[20]

QUESTION 4: LIMITS, FITS AND MANUFACTURING

A shaft must fit in a hole. The hole is Ø25 H7 (+0.021 / 0) and the shaft is Ø25 g6 (−0.007 / −0.020).

4.1 Write down the upper and lower limits of size of the hole and of the shaft. (4)

4.2 Calculate the maximum and the minimum clearance. (4)

4.3 What type of fit is this? (1)

4.4 Name the PRIMARY manufacturing process (casting, forging or extrusion) best suited to each part: (a) a cast-iron engine block, (b) a strong steel spanner, (c) a long aluminium window-frame profile. (3)

4.5 Name the machining (secondary) process used to: (a) make a round hole in the base of the block in FIGURE 1, (b) make a cylindrical shaft on a lathe, (c) cut a flat slot or keyway. (3)

[15]

QUESTION 5: COMPUTER-AIDED DESIGN (CAD)

5.1 Give FOUR advantages of CAD over drawing by hand. (4)

5.2 Explain the difference between 2D drafting and 3D solid modelling, and explain what parametric modelling means. (4)

5.3 Describe, step by step, how you would model the block in FIGURE 1 as a 3D solid using sketches, extrusions and cuts. (6)

5.4 A line starts at the point (20, 30). Give the end point of the line when the second point is entered as:

5.4.1 the relative coordinate @30,40 (2)

5.4.2 the relative polar coordinate @50<90 (2)

5.5 What are STEP and STL files used for? (2)

[20]

TOTAL: 100

END OF QUESTION PAPER